UG (NEP) 5th SEMESTER
APPLIED COMPUTING – THEORY OF COMPUTATION
(ACP522N)

Time Allowed: 2:00 Hours Maximum Marks: 75 Minimum Marks: 27
No. of Questions: 09

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

1. Define finite automaton and mention its basic components.
2. What is the difference between deterministic and non-deterministic finite automata?
3. Write any two closure properties of regular languages.
4. Define Context-Free Grammar (CFG) and mention its main components.
5. What is the difference between recursive and recursively enumerable languages?
6. Give an example of an ambiguous grammar.
7. Define decidability and give one example of a decidable problem.
8. What is the difference between time complexity and space complexity?

SECTION B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Explain the formal definition of computation for a finite automaton with an example.

OR

Write a short note on the equivalence of regular expressions and finite automata.

3. Explain the Chomsky hierarchy of grammars with suitable examples.

OR

Describe the relationship between Deterministic Pushdown Automata (DPDA) and Deterministic Context-Free Grammars (DCFGs).

4. Explain the formal definition of a Turing Machine. Give an example of a simple language accepted by it.

OR

Describe the Halting Problem and explain briefly why it is undecidable.

5. Write short notes on Class P and Class NP with suitable examples.

OR

Construct a finite automaton to recognize all strings over {0,1} that end with “01”.

SECTION C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Prove that for every NFA, there exists an equivalent DFA. Illustrate the conversion process with a suitable example.
7. Explain in detail the closure properties of regular languages under union, concatenation, and Kleene star operations, with examples.
8. Define Pushdown Automata (PDA) formally. Explain the equivalence between PDA and CFG with a suitable example or proof sketch.
9. What is reducibility in Computability Theory? Explain how reducibility helps in proving undecidability of problems, with an example.